

E-Commerce Sales Dataset Analysis

Celebal Summer Internship 2026

Week 2 Task

Objective

The objective of this project is to analyze the Sample Superstore sales dataset using SQL. The analysis includes exploring the dataset, applying filters, performing aggregations, solving business-related queries, and validating data quality to gain useful business insights.

Step 1: Create Database

```
CREATE DATABASE superstore_db;
```

```
USE superstore_db;
```

---> Create Table

The table was created automatically using the **MySQL Table Data Import Wizard** while importing the dataset.

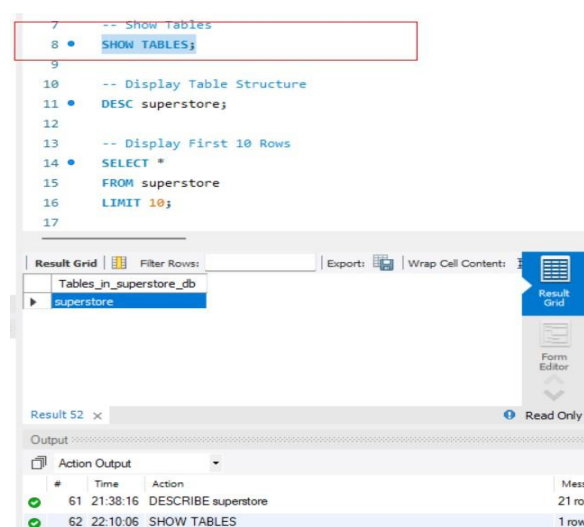
---> Load Dataset

The **SampleSuperstore.csv** dataset was successfully imported into the **superstore** table using the MySQL Table Data Import Wizard.

Step 2: Explore Dataset

```
SHOW TABLES;
```

Output



-- Display Table Structure--

DESC superstore;

Output

SQL File 1* x

Limit to 1000 rows

```

7  -- Show Tables
8  •  SHOW TABLES;
9
10 -- Display Table Structure
11 •  DESC superstore;
12
13 -- Display First 10 Rows
14 •  SELECT *
15  FROM superstore
16  LIMIT 10;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Field	Type	Null	Key	Default	Extra
order_date	text	YES		NULL	
ship_date	text	YES		NULL	
ship_mode	text	YES		NULL	
customer_id	text	YES		NULL	
customer_name	text	YES		NULL	
Segment	text	YES		NULL	
Country	text	YES		NULL	
City	text	YES		NULL	
State	text	YES		NULL	
postal_code	int	YES		NULL	
Region	text	YES		NULL	
product_id	text	YES		NULL	
Category	text	YES		NULL	
sub_category	text	YES		NULL	
product_name	text	YES		NULL	
Sales	double	YES		NULL	
Quantity	int	YES		NULL	
Discount	double	YES		NULL	
Profit	double	YES		NULL	

Result 53 x Read Only

-- Display First 10 Rows--

SELECT * FROM superstore LIMIT 10;

Output

```

13  -- Display First 10 Rows
14 •  SELECT *
15  FROM superstore
16  LIMIT 10;
17
18  -- Count Total Records
19 •  SELECT COUNT(*) AS Total Records

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

row_id	order_id	order_date	ship_date	ship_mode	customer_id
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710

-- Count Total Records--

```
SELECT COUNT(*) AS Total_Records FROM superstore;
```

Output

The screenshot shows a SQL query in a text editor and its execution result in a grid. The query is: `SELECT COUNT(*) AS Total_Records FROM superstore;`. The result grid shows a single row with the value 9694.

Total_Records
9694

Step 3: Apply WHERE filters (region, category, date, sales)

-- Region Filter--

```
SELECT * FROM superstore WHERE region='West';
```

Output

The screenshot shows a SQL query in a text editor and its execution result in a grid. The query is: `SELECT * FROM superstore WHERE region = 'West';`. The result grid shows a list of orders filtered by region 'West'.

row_id	order_id	order_date	ship_date	ship_mode	customer_id
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710
14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070
18	CA-2014-167164	5/13/2014	5/15/2014	Second Class	AG-10270
19	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925
20	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925
21	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870
26	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945
27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945
43	CA-2016-101343	7/17/2016	7/22/2016	Standard Class	RA-19885
63	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720

-- Category Filter--

```
SELECT * FROM superstore WHERE category='Furniture';
```

Output

```

28  -- Category Filter
29  • SELECT *
30  FROM superstore
31  WHERE category = 'Furniture';
32
33  -- Sales Greater Than 1000
34  • SELECT *

```

row_id	order_id	order_date	ship_date	ship_mode	customer_id
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710
24	US-2017-156909	7/16/2017	7/18/2017	Second Class	SF-20065
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870
28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520
30	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520
37	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485
39	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710
40	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710
52	CA-2015-115742	4/18/2015	4/22/2015	Standard Class	DP-13000
53	CA-2015-115742	4/18/2015	4/22/2015	Standard Class	DP-13000
58	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055
66	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720
67	US-2015-164175	4/30/2015	5/5/2015	Standard Class	PS-18970
73	US-2015-134026	4/26/2015	5/2/2015	Standard Class	JE-15745

superstore 55 x Read Only

-- Sales Greater Than 1000--

```
SELECT * FROM superstore WHERE sales>1000;
```

Output

```

33  -- Sales Greater Than 1000
34  • SELECT *
35  FROM superstore
36  WHERE sales > 1000;
37
38  -- Profit Less Than 0 (Loss Orders)
39  • SELECT *
40  FROM superstore

```

row_id	order_id	order_date	ship_date	ship_mode	customer_id
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870
28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520
36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485
55	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265
68	CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590
150	CA-2016-114489	12/5/2016	12/9/2016	Standard Class	JE-16165
166	CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BM-11140
168	CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BM-11140
216	CA-2015-146262	1/2/2015	1/9/2015	Standard Class	VW-21775
245	CA-2014-131926	6/1/2014	6/6/2014	Second Class	DW-13480
248	CA-2014-131926	6/1/2014	6/6/2014	Second Class	DW-13480
252	CA-2016-145625	9/11/2016	9/17/2016	Standard Class	KC-16540
263	US-2014-106992	9/19/2014	9/21/2014	Second Class	SB-20290
264	US-2014-106992	9/19/2014	9/21/2014	Second Class	SB-20290
282	US-2015-161991	9/26/2015	9/28/2015	Second Class	SC-20725
283	CA-2015-130890	11/2/2015	11/6/2015	Standard Class	JO-15280
319	CA-2014-164973	11/4/2014	11/9/2014	Standard Class	NM-18445

superstore 56 x Read Only

-- Profit Less Than 0 (Loss Orders)--

SELECT * FROM superstore WHERE profit < 0;

Output

```

38  -- Profit Less Than 0 (Loss Orders)
39  • SELECT *
40  FROM superstore
41  WHERE profit < 0;
42
43  -- Date Filter

```

	row_id	order_id	order_date	ship_date	ship_mode	customer_id
▶	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335
	15	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815
	16	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815
	24	US-2017-156909	7/16/2017	7/18/2017	Second Class	SF-20065
	28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520
	29	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520
	37	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485
	39	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710
	40	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710
	67	US-2015-164175	4/30/2015	5/5/2015	Standard Class	PS-18970
	73	US-2015-134026	4/26/2015	5/2/2015	Standard Class	JE-15745
	75	US-2015-134026	4/26/2015	5/2/2015	Standard Class	JE-15745
	76	US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600
	79	US-2014-147606	11/26/2014	12/1/2014	Second Class	JE-15745
	85	US-2017-119662	11/13/2017	11/16/2017	First Class	CS-12400
	96	US-2017-109484	11/6/2017	11/12/2017	Standard Class	RB-19705
	101	CA-2016-158568	8/29/2016	9/2/2016	Standard Class	RB-19465
	102	CA-2016-158568	8/29/2016	9/2/2016	Standard Class	RB-19465

superstore 57 x

-- Date Filter--

SELECT * FROM superstore WHERE order_date BETWEEN '2016-01-01' AND '2016-12-31';

Output

```

43  -- Date Filter
44  • SELECT *
45  FROM superstore
46  WHERE order_date BETWEEN '2016-01-01' AND '2016-12-31';
47
48  -- 4.Use GROUP BY for aggregations (sales, quantity, averages)
49  -- Category-wise Sales

```

	row_id	order_id	order_date	ship_date	ship_mode	customer_id
▶	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520
	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520
	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045
	14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070
	22	CA-2016-137330	12/9/2016	12/13/2016	Standard Class	KB-16585
	23	CA-2016-137330	12/9/2016	12/13/2016	Standard Class	KB-16585
	26	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945
	27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945
	36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485
	37	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485
	43	CA-2016-101343	7/17/2016	7/22/2016	Standard Class	RA-19885
	45	CA-2016-118255	3/11/2016	3/13/2016	First Class	ON-18715
	46	CA-2016-118255	3/11/2016	3/13/2016	First Class	ON-18715
	48	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900
	49	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900
	54	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265
	55	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265
	56	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055

superstore 59 x

Step 4: Aggregation

-- Category-wise Sales--

```
SELECT category, SUM(sales) AS Total_Sales FROM superstore GROUP BY category;
```

Output

```
48  -- 4.Use GROUP BY for aggregations (sales, quantity, averages)
49  -- Category-wise Sales
50  • SELECT category,
51      SUM(sales) AS Total_Sales
52  FROM superstore
53  GROUP BY category;
```

	category	Total_Sales
▶	Furniture	733046.8612999996
	Office Supplies	703502.9280000031
	Technology	835900.0669999964

-- Region-wise Average Sales--

```
SELECT region, AVG(sales) AS Average_Sales FROM superstore GROUP BY region;
```

Output

```
55  -- Region-wise Average Sales
56  • SELECT region,
57      AVG(sales) AS Average_Sales
58  FROM superstore
59  GROUP BY region;
```

	region	Average_Sales
▶	South	246.34805889803695
	West	230.2263131655374
	Central	220.2658729203543
	East	243.90205152394708

-- Segment-wise Order Count--

```
SELECT segment, COUNT(*) AS Total_Orders FROM superstore GROUP BY segment;
```

Output

```
61  -- Segment-wise Order Count
62  • SELECT segment,
63      COUNT(*) AS Total_Orders
64  FROM superstore
65  GROUP BY segment;
```

segment	Total_Orders
Consumer	5044
Corporate	2920
Home Office	1730

-- Category-wise Quantity Sold--

```
SELECT category, SUM(quantity) AS Total_Quantity FROM superstore GROUP BY category;
```

Output

```
67  -- Category-wise Quantity Sold
68  • SELECT category,
69      SUM(quantity) AS Total_Quantity
70  FROM superstore
71  GROUP BY category;
```

category	Total_Quantity
Furniture	7855
Office Supplies	21990
Technology	6904

Step 5: Sort and limit results (top products, top categories)

-- Top 10 Products by Sales

```
SELECT product_name, SUM(sales) AS Total_Sales FROM superstore GROUP BY
product_name ORDER BY Total_Sales DESC LIMIT 10;
```

Output

```

73  -- 5.Sort and limit results (top products, top categories)
74  -- Top 10 Products by Sales
75  • SELECT product_name,
76      SUM(sales) AS Total_Sales
77  FROM superstore
78  GROUP BY product_name
79  ORDER BY Total_Sales DESC
80  LIMIT 10;
81  -- Top 10 Customers by Sales

```

product_name	Total_Sales
Canon imageCLASS 2200 Advanced Copier	61599.824
Fellowes PB500 Electric Punch Plastic Comb Bind...	27453.384
Cisco TelePresence System EX90 Videoconferen...	22638.48
HON 5400 Series Task Chairs for Big and Tall	21870.576
GBC DocuBind TL300 Electric Binding System	19823.479000000003
GBC Ibimaster 500 Manual ProClick Binding System	19024.5
Hewlett Packard LaserJet 3310 Copier	18839.686
HP Designjet T520 Inkjet Large Format Printer -...	18374.895
GBC DocuBind P400 Electric Binding System	17965.068
High Speed Automatic Electric Letter Opener	17030.311999999998

-- Top 10 Customers by Sales

```
SELECT customer_name, SUM(sales) AS Total_Sales FROM superstore GROUP BY
customer_name ORDER BY Total_Sales DESC LIMIT 10;
```

Output

```

81  -- Top 10 Customers by Sales
82  • SELECT customer_name,
83      SUM(sales) AS Total_Sales
84  FROM superstore
85  GROUP BY customer_name
86  ORDER BY Total_Sales DESC
87  LIMIT 10;
88

```

customer_name	Total_Sales
Sean Miller	25043.05
Tamara Chand	19017.847999999998
Raymond Buch	15117.339
Tom Ashbrook	14595.62
Adrian Barton	14355.610999999997
Sanjit Chand	14142.333999999999
Ken Lonsdale	14071.917
Hunter Lopez	12873.297999999999
Sanjit Engle	12209.438000000002
Christopher Conant	12129.072

-- Top 5 Cities by Profit

```
SELECT city, SUM(profit) AS Total_Profit FROM superstore GROUP BY city ORDER BY
Total_Profit DESC LIMIT 5;
```

Output

```

89  -- Top 5 Cities by Profit
90  • SELECT city,
91      SUM(profit) AS Total_Profit
92  FROM superstore
93  GROUP BY city
94  ORDER BY Total_Profit DESC
95  LIMIT 5;

```

city	Total_Profit
New York City	61624.0582
Los Angeles	29806.91589999997
Seattle	28868.995700000003
San Francisco	17176.66849999999
Detroit	13117.051600000003

Step 6: Solve use cases (monthly trends, top customers, duplicates).

-- Monthly Sales Trend

```
SELECT MONTH(order_date) AS Month, SUM(sales) AS Total_Sales FROM superstore
GROUP BY MONTH(order_date) ORDER BY Month;
```

Output

```

97  -- 6.Solve use cases (monthly trends, top customers, duplicates).
98  -- Monthly Sales Trend
99  • SELECT MONTH(order_date) AS Month,
100      SUM(sales) AS Total_Sales
101  FROM superstore
102  GROUP BY MONTH(order_date)
103  ORDER BY Month;
104
105  -- Top Customers

```

Month	Total_Sales
NULL	2272449.8562999545

-- Top Customers

```
SELECT customer_name, SUM(sales) AS Total_Sales FROM superstore GROUP BY
customer_name ORDER BY Total_Sales DESC LIMIT 10;
```

Output

```

105      -- Top Customers
106      SELECT customer_name,
107             SUM(sales) AS Total_Sales
108      FROM superstore
109      GROUP BY customer_name
110      ORDER BY Total_Sales DESC
111      LIMIT 10;
112

```

Result Grid | Filter Rows: | Export

	customer_name	Total_Sales
▶	Sean Miller	25043.05
	Tamara Chand	19017.847999999998
	Raymond Buch	15117.339
	Tom Ashbrook	14595.62
	Adrian Barton	14355.610999999997
	Sanjit Chand	14142.333999999999
	Ken Lonsdale	14071.917
	Hunter Lopez	12873.297999999999
	Sanjit Engle	12209.438000000002
	Christopher Conant	12129.072

-- Duplicate Orders

```
SELECT order_id, COUNT() AS Duplicate_Count FROM superstore GROUP BY order_id
HAVING COUNT() > 1;
```

Output

```

113      -- Duplicate Orders
114      SELECT order_id,
115             COUNT(*) AS Duplicate_Count
116      FROM superstore
117      GROUP BY order_id
118      HAVING COUNT(*) > 1;
119

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	order_id	Duplicate_Count
▶	CA-2016-152156	2
	US-2015-108966	2
	CA-2014-115812	7
	US-2015-118983	2
	CA-2014-143336	3
	CA-2016-137330	2
	CA-2016-121755	2
	US-2015-150630	6
	CA-2016-117590	2
	CA-2015-117415	4
	CA-2016-118255	2
	CA-2016-169194	2
	CA-2015-115742	4
	CA-2016-105816	2
	CA-2016-111682	7
	CA-2015-135545	4
	CA-2014-106376	2
	US-2015-134026	3
	US-2017-118038	2
	CA-2016-107300	2

Result 74 x

-- Highest Profit Products

```
SELECT product_name, SUM(profit) AS Total_Profit FROM superstore GROUP BY
product_name ORDER BY Total_Profit DESC LIMIT 10;
```

Output

```

120  -- Highest Profit Products
121  • SELECT product_name,
122      SUM(profit) AS Total_Profit
123  FROM superstore
124  GROUP BY product_name
125  ORDER BY Total_Profit DESC
126  LIMIT 10;
127
128  -- 7 Validate results (row counts, data quality)

```

product_name	Total_Profit
Canon imageCLASS 2200 Advanced Copier	25199.928000000004
Fellowes PB500 Electric Punch Plastic Comb Bind...	7753.039
Hewlett Packard LaserJet 3310 Copier	6983.8836
Canon PC1060 Personal Laser Copier	4570.9347
HP Designjet T520 Inkjet Large Format Printer -...	4094.9766
Ativa V4110MDD Micro-Cut Shredder	3772.9461
3D Systems Cube Printer, 2nd Generation, Mag...	3717.9714000000004
Plantronics Savi W720 Multi-Device Wireless He...	3696.2819999999997
Ibico EPK-21 Electric Binding System	3345.2823
Zebra ZM400 Thermal Label Printer	3343.536

Step 7:Validate results (row counts, data quality)

-- Total Rows--

```
SELECT COUNT(*) AS Total_Rows FROM superstore;
```

Output

```

128  -- 7.Validate results (row counts, data quality)
129  -- Total Rows
130  • SELECT COUNT(*) AS Total_Rows
131  FROM superstore;
132
133  -- Check Null Values

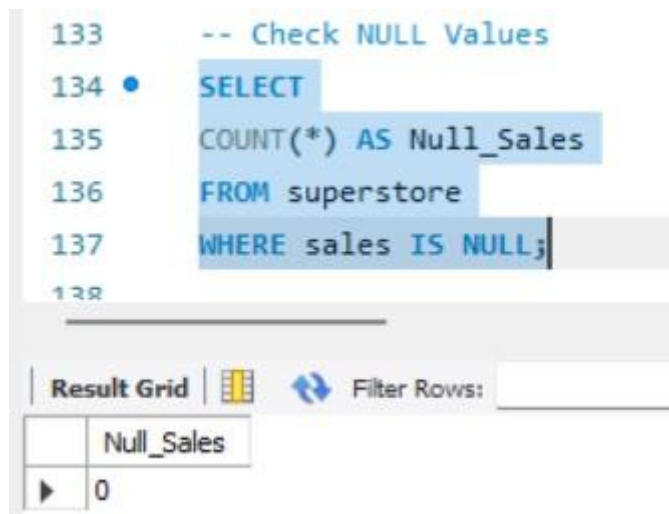
```

Total_Rows
9694

-- Check NULL Values--

```
SELECT COUNT(*) AS Null_Sales FROM superstore WHERE sales IS NULL;
```

Output



The screenshot shows a SQL query editor with the following text:

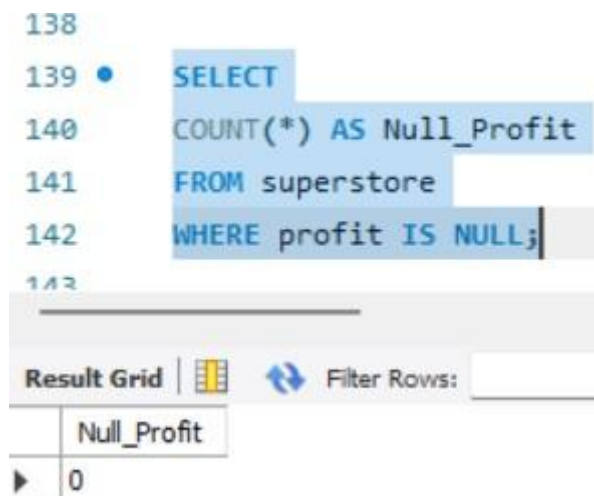
```
133 -- Check NULL Values
134 SELECT
135 COUNT(*) AS Null_Sales
136 FROM superstore
137 WHERE sales IS NULL;
```

Below the editor is a 'Result Grid' section. It includes a 'Filter Rows:' input field and a table with one row and one column:

Null_Sales
0

```
SELECT COUNT(*) AS Null_Profit FROM superstore WHERE profit IS NULL;
```

Output



The screenshot shows a SQL query editor with the following text:

```
138
139 SELECT
140 COUNT(*) AS Null_Profit
141 FROM superstore
142 WHERE profit IS NULL;
```

Below the editor is a 'Result Grid' section. It includes a 'Filter Rows:' input field and a table with one row and one column:

Null_Profit
0

-- Check Duplicate Order IDs--

```
SELECT order_id, COUNT() AS Duplicate_Count FROM superstore GROUP BY order_id
HAVING COUNT() > 1;
```

Output

144 -- Check Duplicate Order IDs

145 • SELECT order_id,

146 COUNT(*)

147 FROM superstore

148 GROUP BY order_id

149 HAVING COUNT(*) > 1;

150

Result Grid

Filter Rows:

Export:

	order_id	COUNT(*)
▶	CA-2016-152156	2
	US-2015-108966	2
	CA-2014-115812	7
	US-2015-118983	2
	CA-2014-143336	3
	CA-2016-137330	2
	CA-2016-121755	2
	US-2015-150630	6
	CA-2016-117590	2
	CA-2015-117415	4
	CA-2016-118255	2
	CA-2016-169194	2
	CA-2015-115742	4
	CA-2016-105816	2
	CA-2016-111682	7
	CA-2015-135545	4
	CA-2014-106376	2
	US-2015-134026	3
	US-2017-118038	2
	CA-2016-127300	2

--> Insights :

- 1. Dataset contains 9994 records.
- 2. West region contributes significantly to total sales.
- 3. Technology products generate high sales.
- 4. Some orders have negative profit.
- 5. Duplicate Order IDs indicate multiple products in a single order.